

SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

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Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SCIFIG-EVAL, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

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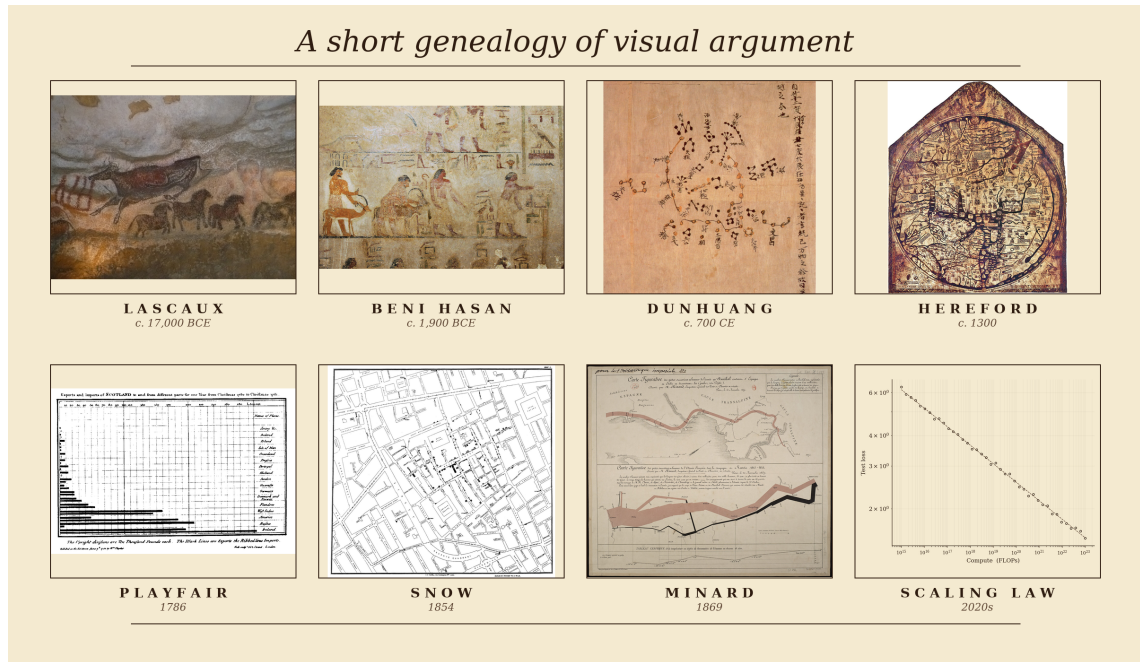
Chapter 1

Introduction

1.1 *Homo sapiens*: A Species of Visual Thinkers

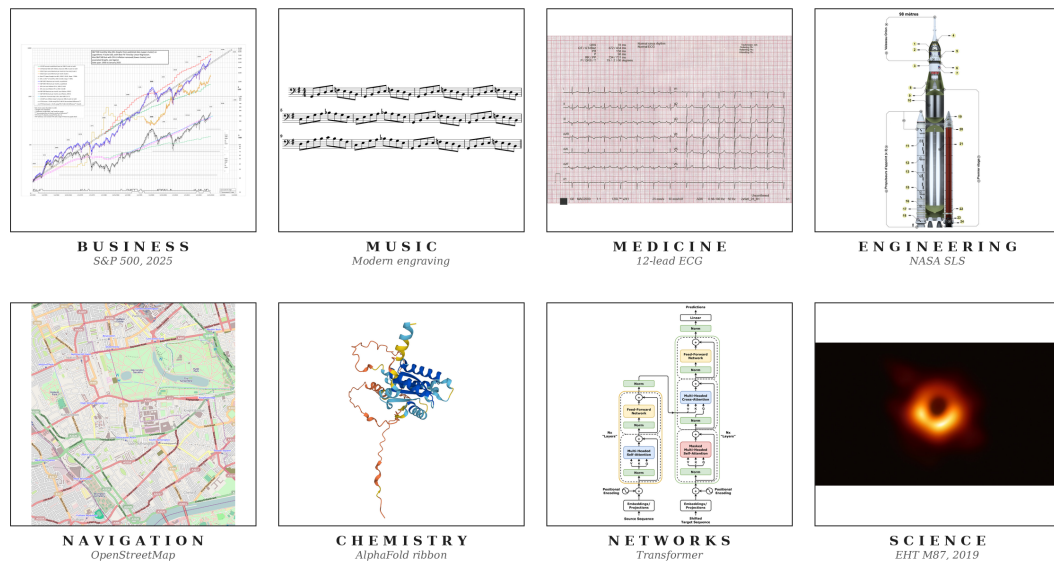
The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.

In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become the lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.



(a)

The figure as the lingua franca of modern knowledge work



(b)

Figure 1.1: (a) A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair’s 1786 bar chart and Snow’s 1854 cholera map to modern neural scaling curves. (b) The figure as lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. *Sources.* (a) Lascaux aurochs, Beni Hasan caravan, Dunhuang star chart, Hereford *mappa mundi*, Playfair 1786 bar chart, Snow 1854 cholera map and Minard 1869 flow map of Napoleon’s Russian campaign, all public domain (via Wikimedia Commons, the Metropolitan Museum of Art, the British Library, Hereford Cathedral, the Internet Archive and the Bibliothèque nationale de France respectively); scaling curves adapted from Kaplan et al. 2020. (b) A candlestick chart, score excerpt, ECG trace, blueprint, nautical chart, molecular diagram, network graph and scientific plot, from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

1.2 Can *Intelligentia Machinae* (Machine Intelligence) Comprehend Visuals Similarly?

Recent progress in artificial intelligence has been uncommonly rapid, and the current generation of large models has brought within plausible reach an older ambition of the field: a machine that can collaborate with a human being to achieve a shared task. A necessary condition of that collaboration is that the two parties reason on similar planes and navigate near-similar cognitive paradigms. Exact cognitive parity is not required; only a proximity close enough to sustain useful work in concert with a human collaborator. Since the human, as the previous section establishes, is a visual thinker, that proximity obliges the machine to comprehend the visual: to read, interpret and reason from the figure with fidelity comparable to its reading of text.

Systems at the current frontier, including Anthropic’s Claude Opus 4.6 and 4.7 (Anthropic, 2026a,b), OpenAI’s GPT-5 series (OpenAI, 2025a) and Google DeepMind’s Gemini 3 Pro (Google DeepMind, 2026a), are now deployed as infrastructure rather than laboratory curiosities. They are shipped into enterprise software through the major cloud providers, and they are measurably altering how code is written and how commercial operations are conducted. The salient question is therefore no longer whether such systems are capable, but whether that capability extends, in the way the human partner’s does, to the visual.

1.3 Motivation for Focusing on Scientific Research Figures

Research is among the first domains in which collaborative intelligence is being tested at scale: between late 2024 and early 2026 the three largest model developers each shipped autonomous research agents (Google, 2024; OpenAI, 2025c; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026d). Yet scientific knowledge is not monolingual; a substantial share of the record is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). Chart- and figure-understanding benchmarks, however, remain predominantly English: recent multilingual efforts such as PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025) and IndicVisionBench (Sharma et al., 2025) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English. The reach of a multimodal system, considered as a research collaborator, therefore scales with the breadth of languages over which its visual reading remains reliable, and the scientific figure, considered across languages, is the natural object at which to test whether the current frontier reads visually or merely appears to. What that reading must withstand, beyond correctness on the clean image, is illustrated in Figure 1.2.

The failure is not a miscount or a mislabelled axis; it is a confident substitution of a plausible category for evidence the model cannot actually see. Honest reading, not just accurate reading, is therefore the object of interest, and the research questions below are framed accordingly: they ask how faithfully today’s frontier models describe scientific figures across languages; how their reading survives visual degradation and adversarial context; and whether, when the evidence is not there, they acknowledge it rather than fill it in.

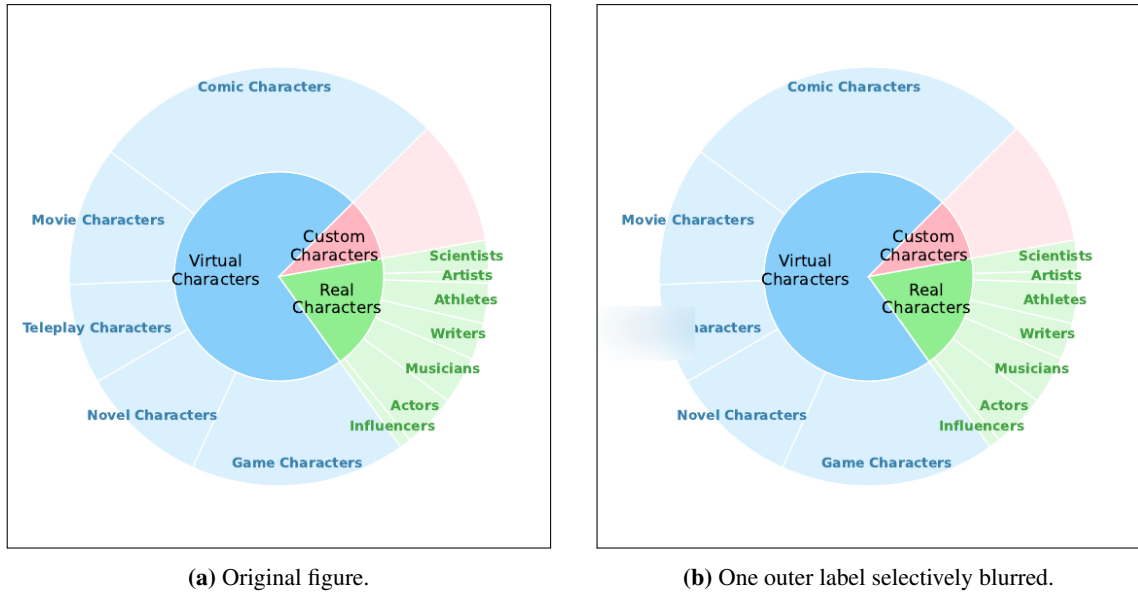


Figure 1.2: A sunburst taken from an arXiv paper in the SCIFIG-EVAL corpus. In (a), the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in (b), that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers “*Anime Characters*,” with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

1.4 Research Questions

The central objective of this thesis is to characterise, across models, languages and adversarial conditions, how faithfully today’s frontier multimodal models actually read scientific figures, and to build an evaluation framework that surfaces the failure modes which conventional accuracy benchmarks obscure. Concretely, the investigation is organised around the following five research questions, each pairing a capacity with the conditions under which that capacity is tested:

RQ1 (Description accuracy across languages). To what extent do current vision–language models produce accurate and complete descriptions of scientific figures across typologically diverse languages?

RQ2 (Robustness to visual degradation). How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy (whether what is stated is correct), completeness (whether what is visible is reported) and clarity (whether what is reported is expressed unambiguously) of VLM-generated figure descriptions?

RQ3 (Comprehension beyond description). Beyond surface-level description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

RQ4 (Adversarial robustness). How robust are VLM outputs when subjected to adversarial manipulations, including misleading captions, contradictory premises and psychologically-informed sycophancy probes?

RQ5 (Calibrated epistemic honesty). When confronted with incomplete or unresolvable visual information, do VLMs exhibit calibrated epistemic honesty, acknowledging uncertainty rather than confabulating plausible but ungrounded responses?

1.5 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Background) provides the technical and conceptual foundation: how frontier multimodal models convert figures into internal representations, how scientific figures carry argument rather than merely depict it, and the failure modes and evaluation traditions on which the remainder draws. Chapter 3 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 4 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 5 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 6 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 7 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 8 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

Chapter 2

Background

This chapter establishes the technical and methodological ground on which the subsequent literature review and evaluation rest. It is organised around four questions about frontier multimodal systems. Section 2.1 asks how today’s Large Language Models see, and describes the architectural pipeline by which an image becomes a representation on which downstream reasoning operates. Section 2.2 asks how they reason about what they see, and how that reasoning behaves in languages other than English. Section 2.3 asks how they behave when the visual evidence is incomplete, misleading or adversarially contested. Section 2.4 asks how the resulting proficiency is measured at scale. The scientific figure as a carrier of argument rather than decoration is taken for granted from the motivation in Chapter 1, and the visualisation-grammar literature is surveyed in Chapter 3.

2.1 How do today’s Large Language Models see?

Contemporary vision-language models are built around a common three-stage pipeline. An image is first partitioned into patches and mapped to a sequence of embeddings by a vision encoder; those embeddings are projected, through a compact connector module, into the token space of a language backbone; and the backbone consumes the projected vision tokens as a prefix alongside the text prompt (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023a). By the time the image reaches the language model it is therefore no longer an image but a sequence of soft tokens drawn from the same vector space as text, with two consequences of first-order relevance to the present work. Any claim the backbone subsequently makes about the figure is, operationally, a claim about these projected tokens rather than about the pixels from which they were derived. And the architectural choices that govern how the tokens are produced are the primary sites at which different model families diverge, and the primary levers through which a family’s visual reading can succeed or fail.

Within this pipeline, four architectural choices distinguish one frontier family from another and turn out to matter empirically. The *vision encoder* is most commonly a Vision Transformer (Dosovitskiy et al., 2021) trained with a contrastive image-text objective of either the CLIP (Radford et al., 2021) or the sigmoid-variant SigLIP (Zhai et al., 2023) form. A growing minority of recent families, led by Qwen2-VL and Qwen3-VL (Bai et al., 2024; Qwen Team, Alibaba, 2025), train a custom encoder from scratch with 2D rotary position embeddings in order to accept images at arbitrary resolution without tiling. The *connector* is either a linear projection into the backbone’s embedding space (Liu et al., 2023), a cross-attention adapter with learnable queries of the

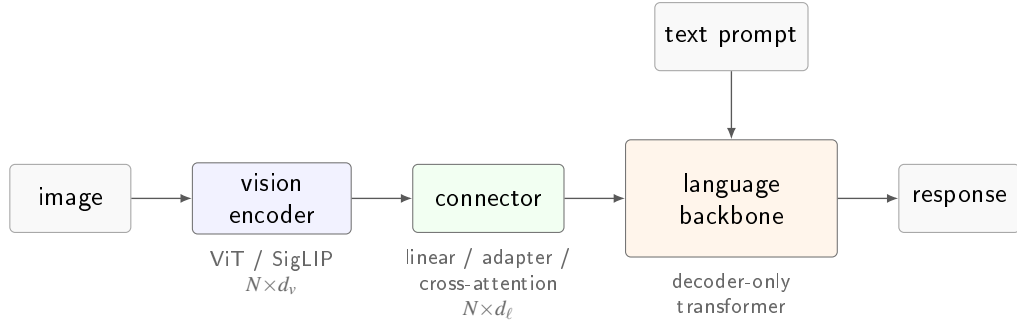


Figure 2.1: The canonical vision-language pipeline. A Vision Transformer (Dosovitskiy et al., 2021), typically trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, maps the image to N patch embeddings of width d_v ; a connector (linear projection, adapter, or cross-attention module) retunes them to the backbone width d_ℓ (Liu et al., 2023; Li et al., 2023a); the decoder-only language model consumes them as a prefix alongside the text prompt.

Table 2.1: Publicly documented vision-language architecture across seven representative families that span the open-weight and closed frontier. “—” denotes a detail that is not disclosed in the primary source. The four open-weight families (top block) publish detailed technical reports, whereas the three closed frontier families (bottom block) publish system cards that address capability and safety but leave the vision stack undisclosed. *Sources.* Gemma 3 (Gemma Team, Google DeepMind, 2025); Qwen3-VL (Qwen Team, Alibaba, 2025), with architectural continuity from Qwen2-VL (Bai et al., 2024); Llama 4 (Meta AI, 2025); Phi-4 multimodal (Microsoft Research, 2025); Claude Opus 4.x (Anthropic, 2026c); GPT-5 / 5.2 (OpenAI, 2025b); Gemini 3 Pro (Google DeepMind, 2026b).

Family	Vision encoder	Resolution / tiling	Image tokens	Fusion regime
Gemma 3	frozen SigLIP-400M	896×896 + Pan&Scan	256 (linear proj.)	adapter
Qwen3-VL	custom ViT + 2D-RoPE	dynamic, any aspect	variable (MRoPE)	native
Llama 4	MetaCLIP-derived ViT	—	— (early-fusion)	native early-fusion
Phi-4 mm	CLIP variant + LoRA	—	—	Mixture-of-LoRAs
Claude Opus 4.x	—	≤ 2576 px long edge	—	—
GPT-5 / 5.2	—	—	—	native (claimed)
Gemini 3 Pro	—	up to 900 images / prompt	—	native (sparse MoE)

Perceiver or Q-Former variety (Alayrac et al., 2022; Li et al., 2023a), or a set of LoRA modules (Hu et al., 2022) spliced directly into the backbone’s weights as in Phi-4 multimodal (Microsoft Research, 2025). The *fusion regime* ranges from a late adapter bolted onto a frozen backbone, through early fusion on interleaved tokens (Meta AI, 2025; Chameleon Team, 2024), to ground-up native co-training of vision and text from initialisation. The *image-token budget*, finally, varies from a fixed allocation per image (Gemma Team, Google DeepMind, 2025) to aspect-preserving variable-length sequences (Bai et al., 2024), and this budget sets a hard upper bound on how much fine-grained detail, from tick labels to legend keys, the encoder can transmit to the backbone in the first place. Table 2.1 summarises where each of the seven representative families sits along these axes.

A second observation about the current frontier follows directly from the table. The open-weight families publish technical reports that document the vision pipeline end to end, whereas the closed frontier families publish only system cards (Anthropic, 2026c; OpenAI, 2025b; Google DeepMind, 2026b) oriented toward capability and safety, and these disclose neither the encoder,

nor the resolution policy, nor the token budget, nor the connector architecture. The models most widely deployed as research infrastructure are therefore the models whose visual pipeline is architecturally least inspectable, and an evaluation that hoped to attribute a failure to a specific pipeline stage by reading internal states would, at the frontier, have almost nothing to read. Behavioural evaluation, conducted purely through controlled perturbations of the input and the text-in, text-out interface every family exposes, is the only regime common to both sides of this asymmetry.

2.2 How do today’s LLMs reason about what they see, and is it impacted by language?

Reading a scientific figure is not one cognitive act but two. *Description* asks what is on the page (the value at a given tick, the colour of a series, the label of a cluster). *Reasoning* asks what follows from it (whether a trend is monotonic, whether two series diverge, whether a conclusion drawn in the accompanying prose is warranted by the plotted data). The two competencies are distinct, and modern chart benchmarks now separate them explicitly. CharXiv (Wang et al., 2024) partitions its evaluation into descriptive and reasoning subsets, and the gap between a model’s scores on the two halves is substantial for every frontier system evaluated. ChartMuseum (Tang et al., 2025), built from harder, naturally occurring charts, reports the same split in sharper form, with the best open-weight model at 38.5%, the best closed model at 63%, and human annotators above 90%.

A natural reading of this gap attributes it to a deficient reasoner. Recent diagnostic work suggests, however, that the bottleneck lies earlier in the pipeline. Before the language backbone can reason about a chart it must first receive, through the connector, an accurate representation of that chart’s perceptual content, and controlled ablations repeatedly show that it is this representation, rather than the reasoning that follows, where most of the error accumulates. Replacing the raw figure with its ground-truth data table, for example, lifts performance sharply even on nominally reasoning-heavy questions (Wang et al., 2024; Tang et al., 2025). Perception, not deduction, is the first-order limit on current chart understanding.

This perceptual bottleneck compounds in multilingual settings. The primary scientific literature is produced in many languages, with substantial shares in Chinese, German, and several Slavic languages, and the consequences of excluding that non-English record for citation, uptake and international collaboration are documented at length (Amano et al., 2023). Chart-understanding benchmarks have not kept pace. The small set of multilingual efforts that do exist (PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025), and IndicVisionBench (Sharma et al., 2025)) each report pronounced cross-lingual degradation on tasks that the same models solve comfortably in English, and PM4Bench further documents that higher-level reasoning scores correlate tightly with lower-level OCR accuracy across languages, consistent with the perception-bottleneck account above.

A non-English figure stresses the pipeline at three distinct layers. At the *pixel* layer, the vision encoder must resolve scripts it was not primarily pretrained on, including Cyrillic diacritics, Chinese logographs at axis-label size, and German compounds that do not fit cleanly into a tight tick space, through the same fixed patch grid the encoder applies to English. At the *token* layer, the backbone’s tokeniser fragments non-English text into finer subword units, which dilutes any lexical prior the backbone has learned and lengthens the sequence over which a claim about the

figure has to be consistently represented. At the *prior* layer, the backbone’s pretraining distribution is itself English-dominant, so that for an underdetermined chart the distribution over continuations that a confident model samples from is systematically skewed toward English-plausible rather than, for example, Bulgarian-plausible content. Script family, morphological complexity, and pretraining density therefore each define a separate axis along which reading can fail, and any serious multilingual evaluation has to stress each of them in turn.

2.3 How do today’s LLMs behave under difficult or adversarial seeing conditions?

Even where they read cleanly on cooperative inputs, frontier vision-language models fail in three recurring ways once the image is degraded, occluded, or framed by misleading text. Each mode is observable only through controlled perturbation of the input, each is compatible with otherwise strong accuracy on clean benchmarks, and each has, in the last two years, attracted dedicated evaluation work.

The first is confabulation when the visual evidence is missing. Asked about a region of an image that has been blurred, cropped, or otherwise rendered illegible, current VLMs routinely produce a confident, plausible-sounding answer in place of an admission of ignorance. Object-level confabulation was first systematically quantified by POPE (Li et al., 2023b); HallusionBench (Guan et al., 2024) extended the measurement to figure-level visual-reasoning traps in which the visual and textual channels are deliberately placed in tension; and ChartHal (Xu et al., 2025) isolates the phenomenon on scientific charts through controlled visual ablations in which the model is interrogated about content the benchmark designers have removed from the image. The mode is already visible on the sunburst of Figure 1.2 in Chapter 1, where GPT-5.2 responds with “Anime Characters” about a blurred outer-ring label as though the label were still legible.

The second is drift under misleading context. *Sycophancy*, the tendency of a language model to adopt an interlocutor’s implied position even when that position is false, was formalised in the text-only setting by Sharma et al. (2023) and has now been documented in the visual setting by Zhao et al. (2024), in which VLMs shift their description of a figure in response to leading questions, misleading captions, or asserted prior beliefs, and sometimes do so in direct contradiction of their own earlier output on the same image. A closely related effect is *textual dominance*, under which the model privileges an assertion stated in the prompt over contrary visual evidence; its signature is the same. In both cases the textual channel systematically overrides what the vision encoder has in fact reported.

The third is confident inference beyond the evidence the image actually supplies. Kadavath et al. (2022) established that, under suitable conditions, language models can know roughly what they know. The visual setting has proved harder. Recent work ties hallucinated output directly to high epistemic uncertainty on early vision-encoder tokens, and to disproportionate attention on a small number of effectively blank patches, so that what appears at the output as careless reasoning is in a significant fraction of cases a quiet perceptual failure at the first layer of the pipeline. On this view, calibration is not a metric bolted onto accuracy but the substrate on which any principled abstention has to rest.

These three failure modes are invisible to any evaluation that scores only the final answer on

a clean image, because all three are compatible with high surface accuracy on such evaluations. Surfacing them requires an evaluation apparatus that probes behaviour rather than correctness, characterising how a model responds when evidence is withdrawn, when it is contradicted by context, and when it is pressed to commit beyond what the image can support. The behavioural framework developed in Chapter 4 is organised along these three lines.

2.4 How do we evaluate the seeing proficiency?

Evaluating visual competence at scale places two distinct demands on the evaluation apparatus. The first is a rubric richer than accuracy on a single-answer question, because the failure modes surveyed in the preceding section coexist with high single-answer accuracy. The second is a grading procedure that can apply such a rubric without a human annotator in the loop for every candidate-prompt pair, because the candidate populations that modern evaluations produce now routinely run to tens of thousands of outputs across models, languages, and perturbation conditions.

The benchmark literature has moved steadily in this direction over the past decade. The first generation of chart-understanding evaluations, PlotQA (Methani et al., 2020) and ChartQA (Masry et al., 2022), paired largely synthetic charts with short-answer questions amenable to exact-match grading. A second generation, including CharXiv (Wang et al., 2024) and ChartQAPro (Masry et al., 2025), moved to real-world scientific charts and separated descriptive from reasoning subsets. The most recent generation, ChartMuseum (Tang et al., 2025) and ChartHal (Xu et al., 2025), has abandoned exact-match grading altogether in favour of open-ended description, multi-step reasoning, and controlled abstention, each of which demands an evaluator capable of judging free-form output against a reference rubric.

For that evaluation, the rubric of choice is Multidimensional Quality Metrics (MQM), introduced by Lommel et al. (2014) for machine translation and now adopted across natural-language generation more broadly. Under MQM, each error in an output is tagged with a category drawn from a fixed typology (accuracy, fluency, terminology, style) and with a severity weight (minor, major, critical), and a calibrated quality score is computed as a weighted aggregation of those annotations. The apparatus transfers directly to figure description, where the typology becomes visual fidelity, completeness, and clarity, and the severity weighting distinguishes a mistaken tick value from a fabricated series.

Human MQM annotation does not, however, scale to the candidate volumes that a modern multimodal evaluation produces. The field has accordingly converged on the *LLM-as-judge* paradigm, first formalised at scale by Zheng et al. (2023), in which a strong general-purpose model grades candidate outputs against a reference rubric. The paradigm is fast, reproducible, and, when deployed carefully, reaches human-level annotation agreement on several evaluation tasks. It is also subject to a well-characterised set of systematic biases. Self-preference, under which a judge rewards outputs that resemble its own family’s style, is documented by Wataoka et al. (2024). Position bias in pairwise comparisons is reported by Zheng et al. (2023) and persists at non-trivial magnitude across most judges. And a family of length, verbosity, and concreteness biases rewards more elaborate output largely irrespective of correctness. The established mitigations, namely randomising candidate position, averaging scores across position swaps, and aggregating across a

heterogeneous ensemble of judges drawn from different families, have been repeatedly validated and now form the methodological baseline for any open-ended VLM evaluation. Together with the behavioural probes of §2.3, they constitute the evaluation apparatus on which Chapters 5 and 6 are built.

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